

Exploring GAN Variants for Balancing Imbalanced Datasets

Special Topics in Data Science
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Dataset: Bank Marketing Dataset

GAN Models: Vanilla GAN and WGAN

Classifier: Random Forest

1. Problem Statement

Class imbalance is a major challenge in supervised machine learning. When one class has significantly fewer samples than the other, predictive models tend to become biased toward the majority class. As a result, the classifier may achieve high overall accuracy while failing to correctly identify minority-class instances.

The purpose of this project is to investigate whether Generative Adversarial Networks (GANs) can be used to generate realistic synthetic samples for the minority class and improve classification performance. Two generative models were implemented: the standard Vanilla GAN and the more advanced Wasserstein GAN (WGAN).

After generating synthetic samples, three classification scenarios were compared:

1. Original imbalanced dataset.
2. Dataset balanced using Vanilla GAN.
3. Dataset balanced using WGAN.

The results were evaluated using multiple metrics including Accuracy, Precision, Recall, F1-score, and AUC-ROC.

2. Dataset Description

The Bank Marketing dataset was obtained from the UCI Machine Learning Repository. It contains customer information collected from a Portuguese banking institution. The objective is to predict whether a customer subscribed to a term deposit.

The target variable is 'deposit':

- no (0): Customer did not subscribe.
- yes (1): Customer subscribed.

The dataset contains both numerical and categorical features such as age, job, marital status, education, balance, housing loan, and contact information.

3. Creating the Imbalanced Dataset

The original dataset was nearly balanced, with:

- no: 5,873 samples
- yes: 5,289 samples

To simulate a real-world class imbalance problem, all majority-class samples were retained while only 1,000 minority-class samples were randomly selected.

Final class distribution:

- Class 0 (no): 5,873 samples
- Class 1 (yes): 1,000 samples

This resulted in a strongly imbalanced dataset suitable for evaluating GAN-based augmentation.

4. Data Preprocessing

Several preprocessing steps were applied before training the GAN models and the classifier.

1. Target Encoding:

The target column was converted into binary labels:

- no $\rightarrow$ 0
- yes $\rightarrow$ 1

2. One-Hot Encoding:

Categorical variables were transformed into dummy variables using `pandas.get_dummies()`.

3. Feature Scaling:

All features were normalized to the range [0, 1] using `MinMaxScaler`.

4. Minority Class Extraction:

Only the 1,000 minority-class samples were used to train the GAN models.

After preprocessing:

- Total samples: 6,873
- Number of features: 42
- Minority samples used for GAN training: 1,000

5. Vanilla GAN Architecture

The Vanilla GAN consists of two neural networks trained in competition.

Generator:

- Input: Random noise vector of dimension 32.
- Dense(64) + LeakyReLU
- Dense(128) + LeakyReLU
- Dense(42) with sigmoid activation

Discriminator:

- Input: 42 features.
- Dense(128) + LeakyReLU
- Dense(64) + LeakyReLU
- Dense(1) with sigmoid activation

Training Configuration:

- Epochs: 1,000
- Batch size: 32
- Optimizer: Adam (learning rate = 0.0002)
- Loss function: Binary Crossentropy

6. Wasserstein GAN (WGAN)

WGAN is an improved GAN variant designed to stabilize training and reduce common issues such as mode collapse.

The discriminator is replaced by a critic, which outputs a real-valued score instead of a probability.

Key differences:

- Wasserstein loss function.
- RMSprop optimizer.
- Weight clipping.
- Multiple critic updates per generator update.

Training Configuration:

- Epochs: 1,000
- Batch size: 32
- Critic updates: 5
- Weight clipping value: 0.01
- Optimizer: RMSprop (learning rate = 0.00005)

7. Synthetic Data Generation

The minority class originally contained 1,000 samples. To balance the dataset, an additional 4,873 synthetic samples were generated using each GAN model.

The resulting balanced datasets were:

Vanilla GAN Balanced Dataset:

- Class 0: 5,873
- Class 1: 5,873

WGAN Balanced Dataset:

- Class 0: 5,873
- Class 1: 5,873

8. Classification Setup

A Random Forest classifier was selected because it performs well on tabular data and handles non-linear relationships effectively.

For each dataset:

- 70% of the data was used for training.
- 30% was used for testing.
- Stratified splitting preserved class proportions.

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- AUC-ROC
- Confusion Matrix

9. Experimental Results

Original Imbalanced Dataset

Confusion Matrix:

```
[[1703, 59],  
 [192, 108]]
```

Metrics:

- Accuracy: 0.8783
- Precision: 0.6467
- Recall: 0.3600
- F1-score: 0.4625
- AUC-ROC: 0.9107

Vanilla GAN Balanced Dataset

Confusion Matrix:

```
[[1711, 51],  
 [198, 1564]]
```

Metrics:

- Accuracy: 0.9293
- Precision: 0.9684
- Recall: 0.8876
- F1-score: 0.9263
- AUC-ROC: 0.9852

WGAN Balanced Dataset

Confusion Matrix:

```
[[1712, 50],  
 [194, 1568]]
```

Metrics:

- Accuracy: 0.9308
- Precision: 0.9691
- Recall: 0.8899
- F1-score: 0.9278
- AUC-ROC: 0.9851

10. Discussion

The original imbalanced dataset produced poor minority-class detection, with recall equal to only 36%. This indicates that most positive cases were missed.

After augmenting the minority class using GAN-generated synthetic samples, recall improved dramatically to approximately 89%. Precision also increased from 64.7% to nearly 97%, and the F1-score doubled from 0.46 to 0.93.

Both GAN models achieved excellent results. WGAN slightly outperformed Vanilla GAN in Accuracy, Precision, Recall, and F1-score, although the difference was small.

These results confirm that GAN-based synthetic data generation is highly effective for addressing class imbalance.

11. Conclusion

This project successfully demonstrated the use of GANs to solve class imbalance in the Bank Marketing dataset.

Key findings:

- GAN-generated synthetic data significantly improved classification performance.
- Recall increased from 36% to nearly 89%.
- F1-score improved from 0.46 to 0.93.
- WGAN achieved the best overall performance.

The study shows that GANs are a powerful alternative to traditional oversampling methods and can substantially enhance predictive modeling in imbalanced classification tasks.

12. Future Work

Possible future extensions include:

- Testing Conditional GAN (CGAN).
- Comparing GANs with SMOTE and ADASYN.
- Applying the methodology to other datasets.
- Exploring deep learning classifiers such as MLP and XGBoost.